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**EMPIRICAL TESTING OF THE FAMA-FRENCH MODEL
USING RUSSIAN STOCK MARKET DATA**

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1. Declaration of Generative AI Use

In the preparation of this Term paper, the authors used Claude (Anthropic), available at <https://claude.ai>, as a generative AI assistant.

The model was applied to the following tasks: translation of complex terminology from Russian into English, grammar checking, improving the academic style and clarity of the written text, and reviewing, optimising, and improving the readability and clarity of Python and R code used for data collection, preprocessing, and factor construction. All substantive content, including the research design, interpretation of results, and formulation of hypotheses and conclusions, was developed independently by the authors.

The use of Claude proved effective for ensuring accuracy in the translation of complex terminology, maintaining a consistent academic tone throughout the paper, and producing clean and well-structured code.

2. Abstract

The present study conducts an empirical test of the Fama–French five-factor model (FF5) on Russian stock market data for the period July 2016 — February 2026. The sample includes 137 Moscow Exchange stocks (116 months of observations). Factors are constructed according to the standard methodology of Fama and French (2015) using independent 2×3 sorts, annual June rebalancing, and value weighting by market capitalization. The market factor is proxied by the IMOEX price index. Model quality is assessed using the Gibbons–Ross–Shanken (GRS) test on 22 test portfolios. To evaluate structural stability, subperiod analysis is conducted with a break point in February 2022.

The GRS test does not reject FF5 on Russian data ($F = 0.719$, $p = 0.809$): none of the 22 portfolios demonstrate significant abnormal returns, while the mean absolute value of alpha is 0.372% per month. Among the factors, CMA demonstrates the highest premium (0.76% per month, $t = 1.91$), followed by SMB (0.72%, $t = 1.47$), while HML is negative (−0.22%) and RMW is effectively zero ($t = 0.03$). Neither factor achieves statistical significance at the 5% level — consistent with the second hypothesis and with evidence for other emerging markets (Foye, 2018; Mosoeu & Kodongo, 2022). Subperiod analysis reveals a structural break after the February 2022 sanctions shock: SMB rises sharply to 1.85% per month ($t = 1.79$), whereas the market premium takes a persistently negative value (−1.30%). This pattern is interpreted as a consequence of the asymmetric impact of sanctions on large export-oriented companies relative to small firms with domestic orientation. Robustness checks on equally weighted portfolios and factors confirm the stability of the main conclusions.

The study contributes to the literature on factor model testing in emerging markets by documenting the transformation of the Russian market's factor structure under the influence of the 2022 sanctions.

Keywords: Fama–French five-factor model, Russian stock market, GRS test, emerging markets, sanctions, factor premiums.

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3. Introduction

3.1. Background

Asset return estimation and the identification of systematic risk factors remain central tasks of empirical finance. The Capital Asset Pricing Model (CAPM), developed by Sharpe (1964) and Lintner (1965), laid the foundational theoretical basis by linking an asset's expected return to a single systematic risk factor — market beta. According to CAPM, only non-diversifiable market risk deserves compensation: a well-diversified investor does not receive a premium for the idiosyncratic risk of an individual company (Sharpe, 1964). Despite its theoretical elegance, the empirical validity of the model proved limited: accumulated evidence revealed persistent anomalies unexplainable by a single-factor structure — in particular, the size effect (Banz, 1981) and the value effect (Rosenberg et al., 1985) — which called into question the adequacy of CAPM as a description of cross-sectional return variation.

In response to these empirical limitations, Fama and French (1993) proposed a three-factor model, supplementing the market factor with two systematic premiums: the size factor (SMB, Small Minus Big) and the value factor (HML, High Minus Low based on the book-to-market ratio). The three-factor model substantially outperformed CAPM in explaining the cross-section of stock returns, incorporating the observed size and value anomalies into a unified risk-oriented structure (Fama & French, 1993). The model quickly became the benchmark specification in empirical asset pricing and spawned an extensive body of international research testing its applicability across various markets and institutional conditions (Mishra & O'Brien, 2019).

Further theoretical development led Fama and French (2015) to expand the three-factor model by including two additional factors — profitability (RMW, Robust Minus Weak) and investment (CMA, Conservative Minus Aggressive) — theoretically grounded through the dividend discount model. On U.S. data, the five-factor model (FF5) demonstrated superior fit quality compared to its predecessor: RMW and CMA captured return patterns that HML could not explain alone (Fama & French, 2015). However, subsequent international tests revealed substantial cross-market variation in the model's efficacy: although Fama and French (2017) confirm that FF5 adequately describes average returns in North America and Europe, they document a persistent anomaly among small, low-profitability companies with aggressive investment policies — the so-called "small growth" problem — which the model is unable to resolve in a number of regions, including Japan, indicating the non-transferability of the factor structure on a universal scale.

3.2. Problem & Gap

Despite compelling results for developed markets, the applicability of FF5 in emerging markets remains debatable. Foye (2018), in a large-scale test of the model on emerging markets, shows that neither the three-factor nor the five-factor version provides convincing asset pricing, while the model's efficacy critically depends on the choice of test portfolios. Mosoeu and Kodongo (2022), testing FF5 using the GMM method on a sample of emerging markets, find that the market factor (MKT) is rejected for the majority of countries, whereas the profitability factor (RMW) unexpectedly proves to be the most informative — a result that fundamentally diverges from the conclusions for developed markets. Taken together, these findings suggest that the factor structure developed on U.S. data does not transfer to emerging markets without substantial adaptation.

The Russian stock market represents a special research interest in this context. It combines features typical of emerging markets and documented in the literature — high concentration of state-owned companies, low liquidity, and dominance of the raw materials sector (Foye, 2018; Mosoeu & Kodongo, 2022) — with institutional peculiarities specific to the post-Soviet environment. These institutional peculiarities call into question the direct applicability of factor structures identified on Western data to the Russian market and require independent empirical verification.

The research is further enhanced in significance by the introduction of large-scale Western sanctions in February 2022. This shock created a unique natural experiment: an opportunity to observe how an external institutional upheaval transforms the structure of factor premiums on an isolated market. The asymmetric impact of sanctions — large exporters and state structures came under direct pressure, whereas small companies with domestic orientation suffered to a significantly lesser degree — theoretically implies a structural shift in the size premium (SMB) not documented in any prior research. At the same time, empirical studies systematically testing FF5 on Russian data using the standard methodology of Fama and French (2015), including the GRS test and subperiod analysis accounting for the sanctions shock, are virtually absent from the academic literature — which defines the research gap of the present work.

3.3. Research Questions & Hypotheses

The present study poses three interrelated questions. The first: is the Fama–French five-factor model capable of adequately describing the return structure of Russian stocks — that is, does it leave statistically significant abnormal returns after controlling for five systematic factors? The second: how did the sanctions shock of February 2022 transform the structure of factor premiums on the Russian market — in particular, did it alter the relative significance of the size factor compared to the other factors in the model? Third: do the profitability and investment factors

achieve statistical significance on the Russian market, consistent with evidence for other emerging markets?

Hypothesis 1. FF5 is not rejected by the GRS test on Russian data: five systematic factors jointly adequately describe the return structure of test portfolios.

Hypothesis 2. The profitability (RMW) and investment (CMA) factors fail to achieve statistical significance at the 5% level, consistent with the pattern documented for emerging markets.

Hypothesis 3. The sanctions shock of February 2022 substantially transforms the structure of factor premiums: the size premium (SMB) sharply intensifies due to the asymmetric impact of sanctions on large and small companies.

The theoretical justification for each hypothesis is developed in detail in Section 4.6, following the literature review.

3.4. Methods Overview and Results Preview

To test the formulated hypotheses, five factors are constructed based on data for 137 stocks of the Moscow Exchange over the period July 2016 — February 2026 (116 months) using the standard methodology of Fama and French (2015): independent 2×3 double sorts with annual June rebalancing and value weighting. As test assets, 25 portfolios formed by quintiles of size and book-to-market ratio are used. Model quality is assessed using the Gibbons–Ross–Shanken test (GRS; Gibbons et al. (1989)), which jointly tests the hypothesis of simultaneous equality to zero of all abnormal returns of test portfolios. The robustness of results is verified on equally weighted portfolios and factors, while subperiod analysis — separately for the periods before and after February 2022 — allows evaluation of the impact of the sanctions shock on the structure of factor premiums.

The main results of the study are as follows. The GRS test does not reject FF5 on Russian data ($F = 0.719$, $p = 0.809$), indicating the absence of systematic abnormal returns, **consistent with** the first hypothesis. Among the five factors, the investment factor (CMA) delivers the highest average premium, narrowly ahead of the size factor (SMB), while the profitability factor (RMW) is effectively zero and the value factor (HML) is negative. Neither the investment nor the profitability factor achieves statistical significance at the 5% level — consistent with the second hypothesis predicting statistical insignificance of both RMW and CMA at the 5% level, in line with evidence for other emerging markets (Foye, 2018; Mosoeu & Kodongo, 2022). Subperiod analysis reveals a sharp strengthening of SMB after February 2022 (1.85%, $t = 1.79$) against the backdrop of a persistently negative market premium (−1.30%), which confirms the third

hypothesis and is interpreted as a consequence of asymmetric sanctions pressure on large and small companies.

3.5. Contribution & Structure

The scientific contribution of the study consists of three aspects. First, a systematic empirical test of FF5 on Russian data is conducted using the standard methodology of the original authors — including the GRS test and subperiod analysis — which fills an evident gap in the emerging markets literature (Foye, 2018; Mosoeu & Kodongo, 2022). Second, the transformation of the factor structure of the Russian market under the influence of the 2022 sanctions is documented for the first time: the sharp strengthening of the size premium is interpreted as a consequence of an asymmetric institutional shock having no direct analogues in the existing literature. Third, robustness checks on equally weighted portfolios and factors confirm that the results are not driven by a few large companies, but reflect a broader pattern across the entire sample. From a practical standpoint, the results are of interest to portfolio managers and institutional investors working with Russian assets: under sanctions pressure, the size factor acquires particular significance in portfolio formation, whereas traditional value-based strategies (HML) do not provide a systematic premium.

The paper is structured as follows. Section 4 contains a review of the relevant literature and justification of the hypotheses. Section 5 describes the data and methodology for constructing the factors. Section 6 presents the results of the regression analysis and the GRS test. Section 7 discusses the obtained results in the context of the existing literature. Section 8 formulates the conclusions, limitations of the study, and directions for future research.

4. Literature Review

4.1. Classical CAPM: Theory and Limitations

The question of how an asset's risk determines its expected return is central to financial theory. The Capital Asset Pricing Model (CAPM), independently developed by Sharpe (1964) and Lintner (1965) as a formal extension of Markowitz's (1952) portfolio theory, proposed an elegant answer: the expected return of any asset depends linearly on its covariance with the market portfolio — the beta coefficient — and is expressed as follows:

$$E(R_i) = R_f + \beta_i \cdot [E(R_m) - R_f] \quad (1)$$

where: $E(R_i)$ — the expected return of asset i ;

R_f — the risk-free rate;

β_i — the measure of systematic risk;

$E(R_m) - R_f$ — the market risk premium.

The theoretical appeal of CAPM is determined by its simplicity: all idiosyncratic risk is diversifiable and therefore not rewarded by the market. This property formally follows from the decomposition of asset return variance:

$$\text{Var}(R_i) = \beta_i^2 \cdot \text{Var}(R_m) + \text{Var}(\varepsilon_i) \quad (2)$$

where: $\beta_i^2 \cdot \text{Var}(R_m)$ — systematic (non-diversifiable) risk;

$\text{Var}(\varepsilon_i)$ — idiosyncratic (diversifiable) risk.

According to CAPM, only the first term is rewarded by the market, and the only source of expected return remains sensitivity to the market factor.

At the same time, beta as a risk measure has a substantial conceptual limitation. In its original definition, beta is expressed through covariance: $\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$, which is equivalent to:

$$\beta_i = \rho(R_i, R_m) \cdot \frac{\sigma_i}{\sigma_m}.$$

Thus, beta represents the product of the asset's correlation with the market and its relative volatility. This means that two assets with the same correlation with the market portfolio but different own volatilities will receive different beta values. Beta simultaneously captures both the direction of the relationship with the market and the asset's specific volatility, which may reflect

other sources of risk — liquidity, financial leverage, operational uncertainty — having no direct relation to systematic market risk in its theoretical sense.

An additional problem is created by the structural relationship between beta and company size. Small companies on average exhibit higher volatility, which mechanically leads to higher beta values through the second multiplier $\frac{\sigma_i}{\sigma_m}$. The consequence is that beta partially serves as a proxy for size: the estimated beta coefficient reflects not only systematic market risk but also the size premium embedded in the return structure. Fama and French (2004) show that when company size is included as an independent variable, the explanatory power of beta substantially decreases — beta loses significance, whereas size remains a significant predictor of returns. This indicates that the historical explanatory power of CAPM was determined to a considerable extent not by the measurement of systematic risk as such, but by the correlation of beta with size.

This thesis is formally confirmed through cross-sectional regression showing a negative dependence of beta on company size: $\beta_i = \delta_0 + \delta_1 \cdot \ln(ME_i) + \xi_i$, $\hat{\delta}_1 < 0$, where: ME_i — market capitalisation of firm i . The negative sign of $\hat{\delta}_1$ implies that larger firms structurally exhibit lower betas, while small firms mechanically obtain higher beta estimates regardless of their true underlying systematic risk.

The consequences for CAPM testing are vividly demonstrated by the Fama–MacBeth regressions (Fama & French, 2004):

Table 1. Cross-Sectional Regressions of Returns on Beta and Size (Stylized Based on Fama and French, 2004)

Model	$\hat{\lambda}_\beta$	$\hat{\lambda}_{size}$	R^2
$E(R_i) = \lambda_0 + \lambda_1 \cdot \beta_i$	positive, insignificant	-	low
$E(R_i) = \lambda_0 + \lambda_1 \cdot \beta_i + \lambda_2 \cdot \ln(ME_i)$	insignificant	negative, significant	higher

In the single-factor model, the coefficient on beta is positive but statistically insignificant — the relationship between beta and return is considerably weaker than CAPM predicts. Upon adding size, the picture changes radically: beta completely loses explanatory power, whereas size remains a significant predictor with a negative sign — smaller size is associated with higher returns. This result directly indicates that beta served as a proxy for size rather than an independent measure of systematic risk.

Thus, CAPM suffers from both a conceptual limitation — the incompleteness of beta as a risk measure — and an empirical one: the persistent anomalies that the model is unable to explain

are the size effect and the value effect (Fama & French, 2004). It was this discrepancy that became the starting point for the development of multifactor asset pricing models.

4.2. The Fama-French Three-Factor Model: Empirical Evidence

In their seminal work, Fama and French (1993) formalized the size and value effects as two additional risk factors, proposing the following three-factor specification:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i \cdot MKT_t + s_i \cdot SMB_t + h_i \cdot HML_t + \varepsilon_{i,t} \quad (3)$$

where: $R_{i,t} - R_{f,t}$ — the excess return of asset i at time t ;

α_i — the intercept (abnormal return);

β_i — the market risk coefficient;

MKT_t — the market risk premium at time t ;

s_i — the SMB coefficient;

SMB_t — the return differential between small- and large-capitalisation portfolios;

h_i — the HML coefficient;

HML_t — the return differential between high- and low-book-to-market portfolios;

$\varepsilon_{i,t}$ — the error term.

Both factors are constructed via independent double sorts on market capitalisation and B/M, using the 30th and 70th percentiles as breakpoints. Using data from NYSE, AMEX, and NASDAQ over 1963–1991, Fama and French demonstrated a substantial improvement in explanatory power relative to CAPM, with adjusted R^2 values for diversified portfolios frequently exceeding 90%.

Subsequent work by Fama and French (1996) expanded the model's evidential base, showing that FF3 absorbs the majority of anomalies documented in the empirical literature up to that time — including the earnings-to-price effect, the cash flow-to-price effect, and long-term return reversals. The authors interpret factor premiums within a risk-oriented paradigm: small companies and value companies are fundamentally riskier, especially during periods of economic downturn, and SMB and HML compensate investors for bearing these additional systematic risks.

Nevertheless, the interpretation of factor premiums remains debated. The behavioral alternative explains premiums through systematic investor errors: overvaluation of growth stocks and undervaluation of value stocks due to cognitive biases, rather than compensation for risk. Jegadeesh et al. (2020) contribute to this debate, showing that tests on individual assets possess greater statistical power than tests on portfolios, yet do not allow a definitive resolution of the

question regarding the nature of premiums. In addition, despite the high explanatory power of FF3, the model is unable to explain a number of persistent patterns — in particular, the profitability premium and the conservative investment effect — which became the theoretical motivation for the development of the five-factor extension.

4.3. The Fama-French Five-Factor Model: theoretical justification and results

Drawing on the theoretical foundation of the Miller and Modigliani (1961) dividend discount model, Fama and French (2015) supplemented the three-factor model with two additional factors, obtaining the following five-factor specification:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i \cdot MKT_t + s_i \cdot SMB_t + h_i \cdot HML_t + r_i \cdot RMW_t + c_i \cdot CMA_t + \varepsilon_{i,t} \quad (4)$$

where: r_i — the RMW coefficient;

RMW_t — the return differential between portfolios of high- and low-profitability firms;

c_i — the CMA coefficient;

CMA_t — the return differential between portfolios of firms with conservative and aggressive investment policies;

remaining variables are as defined in equation (3).

The theoretical motivation for the factors formally follows from the dividend discount model. For a company with constant growth, the current stock price is determined as:

$$P_t = \frac{E_t[Y_{t+1} - \Delta B_{t+1}]}{1+r} \quad (5)$$

where: Y_{t+1} — denotes expected earnings;

ΔB_{t+1} — denotes the change in book value (investment);

r — denotes the expected rate of return.

Two theoretical predictions directly follow from this expression: *ceteris paribus*, a more profitable company has a higher current price and, consequently, a lower expected return; a company with more aggressive investments also has a higher current price and a lower expected return. RMW and CMA capture precisely these patterns in the cross-section of returns.

On U.S. data, FF5 demonstrated superiority over FF3: RMW and CMA captured return patterns that HML could not explain alone. Notably, in the presence of RMW and CMA, the value factor HML becomes largely redundant on U.S. data — its informational content is absorbed by the new factors (Fama & French, 2015).

International tests by Fama and French (2017) confirmed that FF5 generally outperforms FF3 in North America, Europe, Japan, and the Asia-Pacific region, but revealed substantial regional heterogeneity. It is fundamentally important that outside the United States, the HML factor retains significance — a result directly contradicting U.S. data and having immediate implications for testing the model in emerging markets. Mishra and O'Brien (2019) complement this picture from a corporate finance perspective, demonstrating that CAPM and FF5 yield similar cost of equity estimates on average, but systematically diverge for companies with pronounced size and value characteristics — precisely the segment that dominates many emerging markets. The aggregate of these cross-market discrepancies motivates direct empirical testing of the model on emerging markets data.

4.4. The Five-Factor Model in Emerging Markets: Institutional Constraints and Evidence

Emerging markets represent a fundamentally different institutional environment compared to developed markets, on whose data FF5 was calibrated. Lower liquidity, concentrated ownership structures, weak protection of minority shareholders, and less informationally efficient prices can distort the economic mechanisms through which factor premiums are generated. Early works on testing factor models in emerging markets, particularly Cakici et al. (2013), showed that size and value effects are present in most regions, yet their magnitude and statistical significance vary substantially — a result indicating that the factor structure developed on Western data does not transfer to emerging markets without substantial adaptation.

Foye (2018) conducted one of the most comprehensive tests of FF5 on a broad sample of 18 emerging markets covering Asia, Latin America, and Eastern Europe, using the Gibbons–Ross–Shanken test (Gibbons et al. (1989)) as the primary evaluation criterion. The results proved critical regarding the model's applicability: the GRS test rejects FF5 for the majority of markets in all three regions, indicating that significant abnormal returns persist even after controlling for all five factors. At the same time, the market factor and HML remained robustly significant — and notably, HML did not become redundant, as on U.S. data — whereas the profitability and investment factors (RMW and CMA) demonstrated weak and frequently insignificant loadings. Foye (2018) explains this by the unreliability of accounting data in emerging markets, where book

values and reported profitability may not reflect real economic fundamentals, especially in the presence of concentrated state or family ownership.

Mosoeu and Kodongo (2022) confirm these findings using the generalized method of moments (GMM) on an alternative sample of emerging and developed markets. Their results show that profitability is the most important factor for explaining returns in emerging markets in aggregate, yet this conclusion masks substantial cross-country heterogeneity: the market factor is rejected for the majority of individual countries, indicating fundamental structural differences between markets.

Zaremba and Czapkiewicz (2017), focusing directly on emerging markets of Eastern Europe — geographically closest to Russia — demonstrate that anomalies in this region are less effectively captured by standard factor models than in other parts of the world. In particular, the authors show that none of the factor specifications considered provides convincing asset pricing for Eastern European markets, and the explanatory power of the model critically depends on the choice of test portfolios and time horizon. Taken together, the accumulated evidence convincingly indicates that the institutional and structural characteristics of emerging markets systematically undermine the explanatory power of RMW and CMA, whereas HML and the market factor retain their relevance.

4.5. Russian Stock Market: Structural Features and Research Gap

The Russian stock market possesses a set of structural characteristics that distinguish it even against the broad universe of emerging markets. First, a significant share of market capitalization is concentrated in state-controlled enterprises of the energy, metallurgical, and financial sectors. The investment decisions of such companies are determined by political considerations alongside commercial objectives, which directly undermines the economic interpretation of the CMA factor: a company expanding assets in response to a government directive demonstrates fundamentally different behavior than a company reacting to favorable market opportunities. This problem is consistent with Foye's (2018) conclusion that concentrated state ownership substantially distorts the informational content of accounting data on which the profitability and investment factors are constructed.

Second, the sectoral concentration of the Russian market — the dominance of oil and gas and metallurgical sectors in the structure of the Moscow Exchange index — creates a factor structure largely determined by commodity price risk, having no direct analogue within the Fama–French model. Such sectoral concentration is characteristic of a number of emerging markets in Eastern Europe, where, according to Zaremba and Czapkiewicz (2017), standard factor models demonstrate the least explanatory power precisely under conditions of high sectoral concentration.

Third, the geopolitical dimension gives the Russian market unique research status. The sanctions regime introduced in 2022 led to the delisting of Russian stocks from international indices, a sharp reduction in foreign investor participation, and structural disruptions in pricing mechanisms. These shocks create conditions for a structural break in factor premiums, raising the question of the stability of coefficients estimated over the full time horizon relative to pre-sanctions and post-sanctions subperiods. The asymmetric impact of sanctions on large export-oriented companies compared to small firms with domestic orientation theoretically implies a structural shift in the size premium (SMB) — a mechanism consistent with the broader finding of Mosoeu and Kodongo (2022) that systematic risk factors in emerging markets exhibit structural instability across different market conditions.

Despite the obvious relevance of these questions, the empirical literature on factor pricing in the Russian market remains limited. To the best of the authors' knowledge, no published work in a peer-reviewed journal has applied the full FF5 specification to Russian data using the GRS test as the primary evaluation criterion with subperiod analysis accounting for the 2022 sanctions shock. This gap in the literature, in combination with the structural peculiarities outlined above, constitutes the main motivation for the present study.

4.6. Research hypotheses

Drawing on the theoretical and empirical foundations set forth in the preceding sections, the following research hypotheses are formulated.

H1 draws on the results of Foye (2018), who, despite criticism of individual factors, finds no grounds for complete rejection of FF5 in a number of emerging markets. The fact that Russia shares key institutional characteristics of these markets — low liquidity, concentration of state ownership, dominance of the raw materials sector — allows an analogous result to be expected. While the GRS test is a demanding criterion that rejects FF5 for the majority of markets in Foye's (2018) sample, the non-rejection finding holds for a meaningful subset, and Russia's structural profile aligns more closely with that subset than with markets where rejection is documented.

H2 follows from the persistent pattern documented in the emerging markets literature. Foye (2018) shows that RMW and CMA are systematically insignificant across the majority of emerging markets, while Mosoeu and Kodongo (2022), testing the model using GMM on a cross-country sample, confirm that profitability and investment factors behave fundamentally differently than in developed markets. The underlying reason is the low quality of financial reporting and the limited applicability of Western metrics of operating profitability and investment activity to companies operating under non-market incentive structures. By contrast, the size factor (SMB) demonstrates robustness in most emerging markets (Foye, 2018; Cakici et al., 2013), which

motivates the expectation that SMB — rather than RMW or CMA — will constitute the dominant source of cross-sectional return variation on the Russian market.

H3 has no direct analogues in the existing literature — no known study has examined the effect of a sanctions shock on the factor structure of an isolated market. Its economic rationale rests on the asymmetric institutional impact of the February 2022 sanctions: large Russian companies — primarily raw material exporters and state structures — came under direct sanctions pressure, whereas small companies with domestic orientation suffered to a significantly lesser degree. This mechanism is consistent with the broader finding of Mosoeu and Kodongo (2022) that systematic risk factors in emerging markets exhibit structural instability across different market conditions, suggesting that factor premiums are sensitive to shifts in the underlying economic environment.

5. Methods and Data

5.1. Formation of a sample of tickers

The initial securities universe was obtained via the API of the Moscow Exchange Information and Statistical Server (MOEX ISS) from the TQBR trading mode — the primary mode of continuous trading in shares and depositary receipts on the Moscow Exchange. The query returned a complete list of instruments with attributes: ticker identifier, trading status, start date of trading, and lot size. As a result of the initial query, a list of 260 instruments was obtained, to which six filtering criteria were sequentially applied (Table 2), with the fifth criterion — availability of fundamental data — applied at the portfolio construction stage through an inner join with the fundamental data array.

Table 2. Sample Formation Funnel

Step	Criterion	Tickers remaining
1	All securities on the TQBR board	260
2	Active trading status (STATUS = "A")	257
3	Trading commenced no later than 2017	196
4	Missed closing price percentage < 30%	167
5	Availability of fundamental data for constructing factors	144
6	Valid book equity, OP and INV characteristics	137

The first criterion — active trading status — excludes suspended and delisted securities for which market prices are unavailable or unreliable. The second criterion — trading commencement no later than 2017 — ensures a minimum observation horizon sufficient for stable regression estimates in time-series regressions: securities with fewer than 70 months of observations do not allow statistically significant coefficient estimates of the model to be obtained. The lower bound of the sample — July 2016 — corresponds to the period of market stabilization following the completion of the merger of the MICEX and RTS exchanges and the introduction of 2014 sanctions, when the number of regularly traded securities reached a level sufficient for forming representative factor portfolios. The third criterion — a threshold of 30% for missing closing prices — follows standard filtering practice in emerging markets with uneven liquidity (Foye, 2018) and allows exclusion of low-liquidity securities with irregular trading that do not reflect real market valuation.

It should be noted that the applied filtering criteria form a sample with a systematic bias toward more liquid and longer-traded securities: companies with irregular trading and short quotation history are excluded by definition. This limits the representativeness of the sample with respect to small and low-liquidity issuers, but is a standard assumption in empirical research on emerging markets (Foye, 2018; Mosoeu & Kodongo, 2022).

To assess the potential impact of survivorship bias, delisted securities were identified within the initial TQBR universe. Over the sample period July 2016 — February 2026, only three companies that were present in the universe at some point during the sample period subsequently lost active trading status. Given the negligible scale of attrition, survivorship bias is unlikely to materially affect factor premiums or GRS test results.

5.2. Fundamental data

Financial data for companies were obtained from the difan.xyz platform, which aggregates annual reports of Russian issuers based on Moscow Exchange disclosures. For each ticker in the sample, three types of files were downloaded: balance sheet, income statement, and company market value data. The list of extracted variables was determined by the requirements of the Fama and French (2015) methodology for calculating model factors and is presented in Table 3.

Table 3. Fundamental Variables

Source	Variable	Designation	Use
Balance Sheet	Total assets	T.A.	INV Factor
Balance Sheet	Equity	BE	B/M, OP Factor
Income Statement	Revenue	Rev	OP Factor
Income Statement	Cost of sales	COGS	OP Factor
Income Statement	Selling and administrative expenses	SG&A	OP Factor
Income Statement	Interest expense	IntExp	OP Factor
Enterprise Value	Market capitalization	M.E.	B/M, SMB
Enterprise Value	Number of shares outstanding	Shares	ME Recovery

The three datasets were merged into a single dataset by keys `date` and `symbol` using an outer join. The final dataset contains 1,570 observations for 146 unique tickers, corresponding to an average of approximately 10.8 annual observations per company. The intersection of this fundamental data array with the price sample of 167 tickers yields 144 companies (Step 5, Table 2). A further 7 companies are excluded at the factor construction stage by applying three validity conditions: (1) the book-to-market ratio (B/M), operating profitability (OP), and investment activity (INV) characteristics must be non-missing; (2) book equity must be strictly positive ($B/M > 0$); and (3) OP and INV must be finite — observations yielding infinite or undefined values (Inf, NaN) arising from division by zero or missing denominators are discarded. These filtering criteria follow the standard practice of Fama and French (2015) and yield a final sample of 137 stocks.

For a number of tickers, market capitalization values were missing in certain years. Missing values were recovered by calculation: December closing price was multiplied by the number of shares outstanding, with forward/backward fill applied for years without direct observations. After recovery, 28 untreated missing values remained, concentrated predominantly in 2014 and not affecting the regression period (July 2016 — February 2026).

According to the methodology of Fama and French (2015), firm characteristics calculated from year $t-1$ financial statements are applied to returns for the period July t — June $t+1$. This six-month lag ensures guaranteed publication of annual financial reports by the time portfolios are formed and excludes the use of information unavailable to investors on the rebalancing date (look-ahead bias).

In total, 137 unique companies appear in at least one annual sort over the full sample period, with an average of 119 stocks participating in each annual sort.

5.3. Risk-free rate

The key rate of the Bank of Russia, published on the official website of the Central Bank of Russia for the period June 2014 — February 2026, was used as the risk-free rate. The choice of the key rate as a proxy for risk-free return is motivated by its role as the primary instrument of monetary policy, determining the minimum return on short-term ruble instruments without credit risk. This approach corresponds to standard practice in empirical research on the Russian market and is consistent with the general principle of using short-term government rates as the risk-free return (Sharpe, 1964).

The annual rate, expressed as a percentage, was converted to decimal fractions and recalculated into monthly effective yield using the formula: $RF_t = (1 + r_{annual})^{1/12} - 1$, where: r_{annual} — the annual key rate in decimal fractions.

The recalculation is necessary to bring the risk-free rate to the same frequency as stock returns. The obtained RF_t values were used in calculating the excess return of each security as the difference between the monthly stock return and RF_t for the corresponding period.

Three monthly observations (January, November, and December 2020) were missing from the source data and were manually imputed using the Bank of Russia key rate effective during those months: 6.25% per annum for January 2020 and 4.25% per annum for November and December 2020.

5.4. Market factor (MKT)

Data for the Moscow Exchange Index (IMOEX) were obtained via the MOEX ISS API for the period June 2014 — February 2026. From daily closing price values, a monthly series was formed by selecting the last trading day of each month, after which the simple monthly return of the index was calculated: $R_{m,t} = \frac{Index_t - Index_{t-1}}{Index_{t-1}}$.

The market factor was defined as the excess return of the market over the risk-free rate: $MKT_t = R_{m,t} - RF_t$.

The IMOEX price index is used as the market proxy to ensure consistency with individual stock returns, which are also computed from closing prices and do not include dividend income. The use of a total return index in combination with price-only stock returns would introduce a systematic upward bias in the market factor relative to portfolio returns, artificially inflating pricing errors.

5.5. Construction of factors

Factors are constructed according to the methodology of Fama and French (2015) based on independent 2×3 double sorts. At the end of June of each year t , stocks are divided into two groups by market capitalization (Small/Big) relative to the sample median and into three groups by the corresponding characteristic (Low/Neutral/High) relative to the 30th and 70th percentiles. Sorts are conducted independently, yielding six intersecting portfolios for each characteristic: SL, SN, SH, BL, BN, BH. In portfolio notation, the first letter indicates the size group (S — Small, small companies; B — Big, large companies), the second indicates the characteristic group (L — Low, bottom 30%; N — Neutral, middle 40%; H — High, top 30%). Thus, portfolio SH includes small companies with high values of the corresponding characteristic, BL — large companies with low values. Companies with negative book equity are excluded from all sorts (Fama & French, 2015). Characteristics calculated from year $t-1$ financial statements are applied to returns for the period July t — June $t+1$, ensuring a lag of at least six months between the publication of financial reports and portfolio formation.

The book-to-market ratio (B/M) is calculated as the ratio of book equity at the end of year $t-1$ to market capitalization at the end of June of year t : $\left(\frac{B}{M}\right)_{i,t} = \frac{BE_{i,t-1}}{ME_{i,June\ t}}$

Operating profitability is defined as the ratio of operating profit minus interest expense to book equity:

$$OP_{i,t} = \frac{Rev_{i,t} - COGS_{i,t} - SGA_{i,t} - IntExp_{i,t}}{BE_{i,t}} \quad (6)$$

where: $Rev_{i,t}$ — revenue of firm i at time t ;

$COGS_{i,t}$ — cost of goods sold;

$SGA_{i,t}$ — selling, general and administrative expenses;

$IntExp_{i,t}$ — interest expense;

$BE_{i,t}$ — book equity.

Investment activity is defined as the growth rate of total assets: $INV_{i,t} = \frac{TA_{i,t} - TA_{i,t-1}}{TA_{i,t-1}}$. These definitions follow the original specification of Fama and French (2015).

The size factor (SMB) is calculated as the average of three differences between the returns of small and big company portfolios across all three characteristics:

$$\begin{aligned} SMB = & \frac{1}{3} \left[\frac{SL^{BM} + SN^{BM} + SH^{BM}}{3} - \frac{BL^{BM} + BN^{BM} + BH^{BM}}{3} \right] \\ & + \frac{1}{3} \left[\frac{SL^{OP} + SN^{OP} + SH^{OP}}{3} - \frac{BL^{OP} + BN^{OP} + BH^{OP}}{3} \right] \\ & + \frac{1}{3} \left[\frac{SL^{INV} + SN^{INV} + SH^{INV}}{3} - \frac{BL^{INV} + BN^{INV} + BH^{INV}}{3} \right] \quad (7) \end{aligned}$$

where: S — small-capitalisation portfolio; B — large-capitalisation portfolio;

L, N, H — low, neutral, and high terciles respectively;

BM, OP, INV — book-to-market, operating profitability, and investment sorting dimensions.

The value, profitability, and investment factors are calculated as differences between the returns of extreme portfolios:

$$HML = \frac{SH^{BM} + BH^{BM}}{2} - \frac{SL^{BM} + BL^{BM}}{2} \quad (8)$$

where: SH^{BM}, BH^{BM} — small and large high book-to-market portfolios;
 SL^{BM}, BL^{BM} — small and large low book-to-market portfolios.

$$RMW = \frac{SH^{OP} + BH^{OP}}{2} - \frac{SL^{OP} + BL^{OP}}{2} \quad (9)$$

where: SH^{OP}, BH^{OP} — small and large robust (high profitability) portfolios;
 SL^{OP}, BL^{OP} — small and large weak (low profitability) portfolios.

$$CMA = \frac{SL^{INV} + BL^{INV}}{2} - \frac{SH^{INV} + BH^{INV}}{2} \quad (10)$$

where: SL^{INV}, BL^{INV} — small and large conservative (low investment) portfolios;
 SH^{INV}, BH^{INV} — small and large aggressive (high investment) portfolios.

Returns within each portfolio are value-weighted by market capitalization at the end of June and rebalanced annually. Monthly stock returns are winsorized at the 1st and 99th percentiles before factor construction to reduce the influence of extreme observations characteristic of emerging markets with uneven liquidity (Foye, 2018). As test assets, 25 portfolios (5×5) are formed by quintiles of market capitalization and B/M, also value-weighted and rebalanced annually in June.

5.6. Regression specification and GRS test

To assess the pricing power of the model, the following time-dependent regression equation is estimated separately for each of the 25 test portfolios (Fama & French, 2015):

$$R_{i,t} - RF_t = \alpha_i + \beta_i(R_{m,t} - RF_t) + s_i \cdot SMB_t + h_i \cdot HML_t + r_i \cdot RMW_t + c_i \cdot CMA_t + \varepsilon_{i,t} \quad (11)$$

where all variables are as defined in equations (3) and (4).

Regressions are estimated by ordinary least squares (OLS) for the base period July 2016 — February 2026 (116 months). The use of aggregated test portfolios instead of individual stocks

reduces the bias in estimates arising from errors in variables (Jegadeesh et al., 2020) and is standard practice in empirical tests of factor models. Following standard practice in FF5 testing (Fama & French, 2015), test portfolios are formed by independent sorts on size and B/M quintiles, which partially overlap with the factor construction sorts. To mitigate the mechanical relationship between HML and the B/M-sorted test portfolios, factors and test assets are constructed from independent sorts using different breakpoints (median/30th–70th percentiles for factors; quintiles for test portfolios), consistent with the approach of Foye (2018).

The primary criterion for model quality is the GRS test (Gibbons et al. (1989)), which jointly tests the null hypothesis of simultaneous equality to zero of all 25 alphas: $H_0: \alpha_1 = \alpha_2 = \dots = \alpha_{25} = 0$.

The GRS statistic takes the following form:

$$GRS = \frac{T - N - K}{N} \cdot \frac{\hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \hat{\mu}' \hat{\Omega}^{-1} \hat{\mu}} \sim F(N, T - N - K)$$

Where: T — the number of months;

N — the number of test portfolios;

K — the number of model factors;

$\hat{\alpha}$ — the vector of estimated intercepts;

$\hat{\Sigma}$ — the estimate of the residual covariance matrix;

$\hat{\Omega}$ — the estimated factor covariance matrix;

$\hat{\mu}$ — the vector of average factor premiums.

Rejection of H_0 at the 5% significance level indicates the presence of systematic abnormal returns unexplained by the model.

Three portfolios — S2B1, S4B4, and S5B5 — are excluded from all GRS tests (full sample and subperiods) due to chronic underpopulation: these portfolios contain fewer than five stocks in the majority of annual sorts, which renders their return series unreliable for time-series regression. Retaining them would reduce the pre-sanctions subperiod test to $T = 31$ months and $df = T - N - K = 1$, making the GRS statistic numerically unstable. Accordingly, the GRS test is conducted on $N = 22$ portfolios throughout the paper. The excluded portfolios are reported in Table 6 for completeness but are not included in the joint test.

To confirm the reliability of results, all regressions are additionally estimated using an equally weighted (EW) specification, in which both the test portfolios and the factors are recomputed with equal weights across all constituent stocks and compared with the capitalization-

weighted (VW) baseline specification. Equal weights eliminate the dominant influence of large companies and allow assessment of whether the results are reproduced across the entire sample to an equal degree. Subperiod analysis is conducted separately for the periods before February 2022 (67 months) and after (49 months) to evaluate the structural stability of factor premiums under the sanctions shock.

All source code is available in Appendix 1 (Python, data collection) and Appendix 2 (R, statistical analysis).

6. Results

6.1. Descriptive statistics of factors

Table 4 presents descriptive statistics for the five model factors over the base period July 2016 — February 2026 (116 months).

Table 4. Descriptive statistics of factors (% per month)

Factor	Mean	Std	t-stat
MKT	−0.292	6.07	−0.517
SMB	0.723	5.29	1.47
HML	−0.22	3.89	−0.610
RMW	0.011	4.51	0.027
CMA	0.760	4.28	1.91

None of the five factors is statistically significant at the 5% level. Among the factors, CMA demonstrates the highest mean value (0.76% per month), followed closely by SMB (0.72%), while MKT shows the lowest (−0.29%). Standard deviations range from 3.89% (HML) to 6.07% (MKT), reflecting the high volatility of the Russian market over the period under consideration.

6.2. Correlations of factors

Table 5 presents the matrix of pairwise correlations of the five factors of the model for the base period.

Table 5. Factor correlation matrix

	MKT	SMB	HML	RMW	CMA
MKT	1.000	−0.092	−0.004	−0.150	0.076
SMB	−0.092	1.000	−0.082	−0.151	0.360
HML	−0.004	−0.082	1.000	−0.211	−0.091
RMW	−0.150	−0.151	−0.211	1.000	−0.057
CMA	0.076	0.360	−0.091	−0.057	1.000

Correlations between factors are generally low. The highest correlation is observed between SMB and CMA (0.360). The negative correlation between HML and RMW is −0.211. Correlations of the remaining factor pairs do not exceed 0.151 in absolute value, indicating relative independence of factors and low risk of multicollinearity in regression estimates.

6.3. Alphas of test portfolios

Table 6 presents estimates of abnormal returns (alphas) for 25 test portfolios formed by quintiles of size and book-to-market ratio. Portfolios are denoted as SiBj, where i is the size quintile (S1 — smallest companies, S5 — largest), j is the B/M quintile (B1 — lowest value, B5 — highest).

Table 6. Alphas of 25 5×5 portfolios (% per month, VW)

Port	α (%/month)	t-stat	p-value	Adj.R ²	Sig
S1B1	−0.001	0.00	0.999	0.579	
S1B2	0.921	0.99	0.323	0.300	
S1B3	0.579	1.10	0.272	0.741	
S1B4	0.081	0.19	0.850	0.795	
S1B5	1.040	1.37	0.173	0.362	
S2B1	0.187	0.24	0.809	0.497	
S2B2	−0.026	−0.05	0.961	0.720	
S2B3	0.654	0.80	0.427	0.606	
S2B4	0.677	1.27	0.206	0.518	
S2B5	0.102	0.28	0.782	0.709	
S3B1	0.185	0.35	0.730	0.574	
S3B2	0.349	0.63	0.528	0.597	
S3B3	0.397	0.61	0.545	0.672	
S3B4	0.455	0.78	0.436	0.490	
S3B5	0.485	1.46	0.148	0.768	
S4B1	−0.407	−0.78	0.440	0.530	
S4B2	−0.316	−0.90	0.368	0.746	
S4B3	0.042	0.12	0.902	0.649	
S4B4	−0.005	−0.01	0.993	0.583	
S4B5	−0.165	−0.29	0.771	0.485	
S5B1	0.211	0.66	0.508	0.742	
S5B2	0.287	0.72	0.474	0.765	
S5B3	0.672	1.95	0.053	0.790	*
S5B4	0.137	0.32	0.752	0.623	
S5B5	0.528	0.94	0.349	0.577	

Note: * — significance at 10%; ** — significance at 5%; *** — significance at 1%; All 25 portfolios are reported for descriptive purposes. Portfolios S2B1, S4B4, and S5B5 are excluded from the GRS test due to chronic underpopulation (fewer than 5 stocks in the majority of annual sorts); see Section 5.6.

None of the 22 portfolios demonstrates a statistically significant alpha at the 5% level. The mean absolute value of alpha across all portfolios is 0.372% per month. Adjusted R² values range from 0.300 (S1B2) to 0.795 (S1B4), with the lowest values observed for portfolios of small companies with low B/M. Portfolio S5B3 approaches significance at the 10% level ($\alpha = 0.67\%$, $t = 1.95$, $p = 0.053$). The significance threshold adopted throughout this paper is 5%. Portfolio S5B3

is flagged at the 10% level in Table 6 for descriptive purposes only and is not treated as evidence against the model.

6.4. Gibbons–Ross–Shanken test (GRS test)

Table 7 presents the results of the Gibbons–Ross–Shanken test, which jointly tests the null hypothesis of simultaneous equality to zero of all 22 alphas of the test portfolios.

Table 7. GRS test results

Statistics	Value
GRS F-statistic	0.719
p-value	0.809
Mean $ \alpha $ (% per month)	0.372%
Number of portfolios (N)	22
Number of months (T)	116
Number of factors (K)	5

Three chronically underpopulated portfolios (S2B1, S4B4, S5B5) are excluded from the GRS test, reducing N to 22 and ensuring a consistent set of test assets across the full sample and subperiods (see Section 6.5). At a significance level of 5%, the null hypothesis is not rejected ($F = 0.719$, $p = 0.809$). The average absolute value of alpha is 0.372% per month.

6.5. Subperiod analysis

Table 8 presents descriptive statistics of factor premiums separately for two sub-periods: before February 2022 (67 months) and after (49 months).

Table 8. Factor premiums by sub-periods (% per month)

Factor	Until 2022 (mean)	Until 2022 (t)	After 2022 (mean)	After 2022 (t)
MKT	0.445	0.84	−1.300	−1.17
SMB	−0.102	−0.28	1.850	1.79
HML	−0.195	−0.43	−0.254	−0.43
RMW	0.007	0.01	0.018	0.03
CMA	0.725	1.64	0.808	1.11

In the subperiod before February 2022, all five factor premiums are statistically insignificant. MKT is positive (0.445%, $t = 0.84$), reflecting moderate positive market returns in the pre-sanctions period. SMB, HML, and RMW are close to zero, while CMA shows a positive but insignificant premium (0.725%, $t = 1.64$).

In the subperiod after February 2022, the picture changes. MKT turns negative (−1.300%, $t = -1.17$), reflecting the overall market decline under the influence of sanctions and rising geopolitical uncertainty. SMB rises sharply to 1.850% per month ($t = 1.79$) — the highest value among all factors in this subperiod. CMA increases to 0.808% ($t = 1.11$). HML remains negative

(−0.254%), RMW remains near zero (0.018%). None of the factors achieves statistical significance at the 5% level in either subperiod.

Table 9. GRS test results by subperiods (N=22 portfolios)

Subperiod	T	N	df	GRS F	p-value	Mean α (%)
Full sample (Jul 2016 – Feb 2026)	116	22	89	0.719	0.809	0.372
Pre-sanctions (Jul 2016 – Jan 2022)	67	22	40	0.930	0.562	0.616
Post-sanctions (Feb 2022 – Feb 2026)	49	22	22	0.968	0.530	0.578

Note: Pre- and post-sanctions subperiod tests are conducted on 22 portfolios, excluding S2B1, S4B4, and S5B5, which remain unpopulated prior to July 2019 due to insufficient sample size. Retaining all 25 portfolios would require restricting the pre-sanctions subperiod to the interval during which all portfolios are populated ($T \approx 31$ months), yielding $df = T - N - K = 31 - 25 - 5 = 1$, which renders the GRS statistic unreliable. Reducing to 22 portfolios allows the full pre-sanctions window ($T = 67$ months) to be used, raising df to 40.

To formally assess whether FF5 maintains its pricing adequacy within each subperiod, GRS tests are conducted separately for the pre- and post-sanctions periods on a reduced set of 22 portfolios (Table 9). Three portfolios — S2B1, S4B4, and S5B5 — are excluded as they remain unpopulated prior to July 2019, which would reduce the pre-sanctions test to $T=31$ months and $df=1$, rendering the statistic unreliable. The GRS test does not reject FF5 in either subperiod: $F=0.930$ ($p=0.562$) before February 2022 and $F=0.968$ ($p=0.530$) after. Mean absolute alpha is 0.616% and 0.578% per month respectively — somewhat higher than the full-sample estimate of 0.372%, consistent with shorter estimation windows and noisier coefficient estimates.

6.6. Robustness check: equally weighted portfolios and factors

Table 10 presents a comparison of test portfolio alphas for the baseline value-weighted specification (VW) and the alternative equally weighted specification (EW).

Table 10. Comparison of VW and EW specifications

Port	α VW (%)	t VW	α EW (%)	t EW
S1B1	-0.001	0.00	0.019	0.04
S1B2	0.921	0.99	0.570	0.78
S1B3	0.579	1.10	0.323	0.67
S1B4	0.081	0.19	-0.216	-0.54
S1B5	1.040	1.37	0.316	0.55
S2B1	0.187	0.24	0.247	0.33
S2B2	-0.026	-0.05	-0.366	-0.76
S2B3	0.654	0.80	0.291	0.41
S2B4	0.677	1.27	0.335	0.66
S2B5	0.102	0.28	-0.251	-0.76
S3B1	0.185	0.35	0.572	1.17
S3B2	0.349	0.63	0.249	0.48
S3B3	0.397	0.61	0.393	0.57
S3B4	0.455	0.78	0.006	0.01
S3B5	0.485	1.46	0.276	0.81
S4B1	-0.407	-0.78	-0.339	-0.67
S4B2	-0.316	-0.90	-0.194	-0.55
S4B3	0.042	0.12	0.140	0.37
S4B4	-0.005	-0.01	-0.189	-0.33
S4B5	-0.165	-0.29	-0.468	-0.85
S5B1	0.211	0.66	0.216	0.74
S5B2	0.287	0.72	0.043	0.13
S5B3	0.672	1.95	0.701	2.13 **
S5B4	0.137	0.32	-0.511	-1.30
S5B5	0.528	0.94	0.372	0.70

Note: * — significance at 10%; ** — significance at 5%; *** — significance at 1%.

Upon switching to equal weights, the mean absolute value of alpha slightly decreases from 0.372% to 0.304%, and the number of significant portfolios increases from 0 to 1 at the 5% level (S5B3, $p = 0.035$). Portfolio S5B3 achieves significance at the 5% level in the EW specification (t

= 2.13, $p = 0.035$) and approaches significance in VW ($t = 1.95$, $p = 0.053$). The GRS test does not reject the model in either specification, indicating that the results are not driven by the capitalization-weighting scheme.

7. Discussion

7.1. Main result: GRS test

The main result of the study is that FF5 is not rejected by the GRS test on Russian stock market data ($F = 0.719$, $p = 0.809$), which is **consistent with** the first hypothesis. This result indicates that five systematic factors jointly adequately describe the return structure of 25 test portfolios, leaving no statistically significant abnormal returns at the system-wide level. Of the 22 portfolios, none demonstrates a statistically significant alpha at the 5% level, providing no evidence against the adequacy of FF5 as a pricing model for Russian stocks.

The obtained result is consistent with the conclusions of Foye (2018), who, despite criticism of individual factors in emerging markets, finds no grounds for complete rejection of FF5 in a number of samples. At the same time, the mean absolute value of alpha is 0.372% per month, indicating a moderate residual of unexplained return and consistent with the general picture for emerging markets, where the model describes returns less precisely than in developed markets (Fama & French, 2017).

7.2. Individual factors and hypothesis 2

Despite the fact that the model as a whole is not rejected, none of the five factors is statistically significant at the 5% level, **consistent with** the second hypothesis. Notably, RMW is effectively zero ($t = 0.03$), while CMA demonstrates the strongest loading among all factors ($t = 1.91$), surpassing even SMB ($t = 1.47$) yet still falls short of the 5% significance threshold. This pattern is typical for emerging markets with limited liquidity and short time series (Foye, 2018; Mosoeu & Kodongo, 2022).

7.2.1. Market factor (MKT)

The market premium is negative on average over the entire period (-0.29% per month, $t = -0.52$), which reflects the fact that IMOEX is a price-only index and does not account for dividend income. The true total return to equity investors is higher, as Russian blue-chip companies historically distribute 7–12% per annum in dividends, implying a downward bias of approximately 0.6–1.0% per month — sufficient to shift the sign of MKT from negative to positive under realistic dividend assumptions. Critically, since both the market proxy and individual stock returns are computed from closing prices without dividend adjustment, the bias enters symmetrically into MKT and all portfolio excess returns. As a result, the estimated factor loadings on MKT (β coefficients) remain unaffected, and the GRS test — which evaluates the joint significance of pricing errors relative to the mean-variance frontier spanned by the factors — is not invalidated by a uniform downward shift in absolute return levels. The limitation therefore affects the

interpretation of the market premium in isolation but does not compromise the cross-sectional pricing results. Mosoeu and Kodongo (2022) also document rejection of the market factor for a number of emerging markets, which is consistent with the observed instability of MKT across subperiods.

7.2.2. Size factor (SMB)

SMB demonstrates one of the highest average premiums among all five factors (0.72% per month, $t = 1.47$), indicating the presence of a small-firm effect on the Russian market. While SMB itself does not reach formal statistical significance at the 5% level, it demonstrates one of the two highest premiums alongside CMA (0.76%), supporting the directional prediction of H2. This result is consistent with the conclusions of Foye (2018), who documents the robustness of SMB in most emerging markets, and agrees with the theoretical rationale of Fama and French (1993): small companies bear additional systematic risks — lower liquidity, limited access to capital, and higher sensitivity to macroeconomic shocks — for which investors demand compensation.

7.2.3. Value factor (HML)

HML is negative (−0.22%), indicating the absence of a classical value premium on the Russian market over the period under consideration. This result is not exceptional for emerging markets: Fama and French (2017) document substantial variation in the value effect across regions, while Foye (2018) finds that HML behaviour in emerging markets systematically differs from U.S. patterns, particularly where concentrated state or family ownership causes book values to diverge from real economic fundamentals. For the Russian market specifically, two structural mechanisms may explain the negative HML. First, the largest Russian companies — predominantly state-controlled energy and metallurgical exporters — exhibit low book-to-market ratios due to accumulated sanctions discounts and political risk premia embedded in their market valuations, yet deliver systematically weak returns precisely because of these risks; they thus occupy the value portfolio while underperforming. Second, the sample period 2016–2026 captures the rapid growth of domestically oriented technology and consumer companies with low B/M ratios — including Yandex and Sber in its technology capacity — which substantially outperformed commodity-sector value stocks, mechanically reversing the sign of HML. Together, these mechanisms suggest that the negative value premium reflects the specific sectoral and institutional structure of the Russian market rather than a general failure of the value concept.

7.2.4. Profitability and investment factors (RMW and CMA)

RMW and CMA are positive but statistically insignificant ($t = 0.03$ and $t = 1.91$, respectively). The weakness of these factors in emerging markets is one of the key findings of Foye (2018) and is explained by low quality of financial reporting, lack of transparency in corporate governance, and limited applicability of Western metrics of operating profitability and investment activity to companies with non-market incentives. Jegadeesh et al. (2020) also point out that the explanatory power of factors may be determined by correlation with observed firm characteristics rather than independent risk premiums, which is especially relevant for markets with a high share of state participation.

The moderate positive correlation between SMB and CMA (0.360, Table 5) warrants attention in interpreting the CMA premium. Small companies in the Russian market tend to exhibit conservative investment behaviour — partly due to limited access to capital markets and partly due to the dominance of domestically oriented firms with low capital expenditure needs. This structural overlap means that part of the CMA premium may reflect the size effect rather than an independent investment factor. While the independent sorting procedure ensures that SMB and CMA are constructed from separate sorts, the residual correlation suggests that their economic content partially overlaps on the Russian market — a pattern consistent with Fama and French (2015), who document a similar SMB–CMA relationship on U.S. data, and which reinforces the interpretation of CMA as a secondary size-related factor rather than a purely independent pricing dimension in this institutional context.

7.3. Sanctions shock and hypothesis 3

Subperiod analysis reveals substantial structural instability of factor premiums caused by the sanctions shock of February 2022, which confirms the third hypothesis. Before February 2022, all factor premiums are statistically insignificant and close to zero, consistent with the general picture for emerging markets under relatively stable macroeconomic conditions (Foye, 2018).

Transformation of the market premium. After February 2022, the market premium turns negative (-1.300% , $t = -1.17$), reflecting the overall market decline under the influence of sanctions, the freezing of non-resident assets, and rising geopolitical uncertainty. This result is consistent with the broader finding of Mosoeu and Kodongo (2022) that the market factor exhibits particularly weak and unstable behaviour in emerging markets across different economic conditions.

Strengthening of SMB. The most significant change is the sharp strengthening of the size factor: SMB rises from -0.102% to 1.850% per month ($t = 1.79$). The economic interpretation of this result is as follows. Large Russian companies — primarily raw material exporters and state

structures — came under direct sanctions pressure: export restrictions, disconnection from international payment systems, and the departure of foreign partners disproportionately hit the largest issuers. Small companies with predominantly domestic business orientation suffered to a significantly lesser degree, which generated a systematic premium for small size in the post-sanctions period. This mechanism has no direct analogues in the existing literature — none of the reviewed works have examined the effect of a sanctions shock on the factor structure of an isolated market — which defines the independent scientific contribution of the present result.

Strengthening of CMA. CMA also increases after 2022 (0.808%, $t = 1.11$): under conditions of high uncertainty and rising cost of capital, companies with conservative investment policy demonstrated relatively higher returns. This result is consistent with the theoretical logic of Fama and French (2015): in periods of heightened uncertainty, aggressive investments carry additional risk, for which the market demands compensation.

HML and RMW. The value and profitability factors show no significant changes between subperiods: HML remains negative in both subperiods (−0.195% and −0.254%), RMW remains near zero, edging slightly from 0.007% to 0.018%. This result indicates that the sanctions shock primarily transformed the size component of the factor structure, whereas the value and profitability components proved less sensitive to institutional changes.

Importantly, the subperiod GRS tests (Table 9) reveal that the structural transformation of factor premiums documented above did not undermine the overall pricing adequacy of FF5: the model is not rejected in either subperiod, with p-values of 0.562 and 0.530 respectively. This finding suggests that while the sanctions shock substantially altered the *magnitude* of individual factor premiums — most notably SMB — the five-factor structure remained sufficient to span the mean-variance frontier of test portfolios in both institutional regimes. The near-identical GRS statistics across subperiods (0.930 vs. 0.968) further indicate that the model's pricing errors did not systematically worsen in the post-sanctions period despite the structural break in factor premiums.

7.4. Robustness of results

Robustness checks on equally weighted portfolios and factors reveal that the main conclusions remain qualitatively intact. Switching to equal weights slightly decreases the mean absolute value of alpha from 0.372% to 0.304%, and one portfolio (S5B3) achieves significance at the 5% level in the EW specification ($p = 0.035$). The GRS test does not reject the model in either specification, confirming that FF5 adequately describes the overall return structure regardless of the weighting scheme.

The differences between VW and EW specifications are informative. Portfolio S5B3 achieves significance at the 5% level in EW ($t = 2.13$) and approaches significance in VW ($t =$

1.95), suggesting that the mild pricing error in this portfolio reflects a genuine pattern rather than a capitalization-driven artifact.

Taken together, the robustness check results are consistent with standard practice in verifying factor models (Fama & French, 2015) and indicate that the study's main conclusions are not an artifact of a specific methodological specification.

8. Conclusion

The present study conducts a systematic empirical test of the Fama–French five-factor model on Russian stock market data for the period July 2016 — February 2026, using a sample of 137 Moscow Exchange stocks. Factors are constructed according to the standard methodology of Fama and French (2015) via independent 2×3 sorts with annual June rebalancing and value weighting. Model quality is evaluated using the GRS test on 22 test portfolios, and subperiod analysis — with a structural break point at February 2022 — allows direct assessment of how the sanctions shock transformed the factor premium structure of an isolated market. Together, these methodological choices allow the three research hypotheses to be tested in a unified and internally consistent framework.

The main result is that FF5 is not rejected by the GRS test on Russian data ($F = 0.719$, $p = 0.809$), with none of the 22 test portfolios exhibiting statistically significant abnormal returns at the 5% level. This places Russia among the minority of emerging markets where the five-factor structure passes the joint pricing test — a result consistent with H1 and with the subset of markets in Foye's (2018) sample where FF5 is not rejected. Despite the institutional peculiarities of the Russian market — concentrated state ownership, sectoral dominance of raw material exporters, and limited liquidity — five systematic factors are jointly sufficient to span the mean-variance frontier of the test assets. The mean absolute alpha of 0.372% per month indicates a moderate residual of unexplained return, consistent with the general picture for emerging markets where the model describes returns less precisely than in developed markets (Fama & French, 2017).

The factor premium structure of the Russian market broadly mirrors the pattern documented for other emerging markets, confirming H2. Size (SMB, 0.72% per month) and investment (CMA, 0.76%) are the dominant sources of cross-sectional return variation, while profitability (RMW) is effectively zero and value (HML) is negative. The statistical insignificance of all five factors — including RMW and CMA — reflects the combination of high return volatility, a relatively short time series, and the structural distortions in accounting data characteristic of markets with concentrated state ownership, rather than the absence of underlying economic mechanisms. The moderate positive correlation between SMB and CMA (0.360) further suggests that these two factors partially share economic content on the Russian market, with small firms tending toward conservative investment behaviour due to limited capital market access — a pattern that reinforces the interpretation of CMA as a secondary size-related dimension in this institutional context.

Subperiod analysis provides the most novel result of the study, confirming H3. The February 2022 sanctions shock constitutes a rare natural experiment in factor pricing: an abrupt, externally imposed institutional disruption that transformed the factor structure of an isolated

market within a single calendar month. The sharp strengthening of SMB — from -0.10% before February 2022 to 1.85% per month after ($t = 1.79$) — is directly traceable to the asymmetric exposure of large exporters and state structures relative to small domestically oriented firms. This mechanism has no direct analogue in the existing literature and represents an independent contribution of the present study. Importantly, this structural shift did not undermine the overall pricing adequacy of FF5: the model is not rejected in either subperiod, with GRS p-values of 0.562 and 0.530 respectively, suggesting that the five-factor structure is robust to institutional regime changes of this magnitude.

The consistency of results across value-weighted and equally weighted specifications confirms that the findings reflect broad market-wide patterns rather than the mechanical influence of a small number of large-capitalisation stocks, further strengthening confidence in the robustness of the main conclusions.

The study contributes to the literature on empirical testing of factor models in emerging markets in three respects. First, it provides the first systematic test of FF5 on Russian data using the standard Fama–French methodology, the GRS test, and subperiod analysis — filling an evident gap identified by Foye (2018) and Mosoeu and Kodongo (2022). Second, it documents the transformation of the Russian market's factor structure under the influence of the 2022 sanctions, offering a novel interpretation of how external institutional shocks reshape systematic risk premiums on an isolated market. Third, robustness checks on equally weighted portfolios confirm that the results are not an artifact of capitalization weighting. From a practical standpoint, the results indicate that under sanctions pressure the size factor acquires particular significance in portfolio formation, whereas traditional value-based strategies do not provide a systematic premium on the Russian market — a finding directly relevant to portfolio managers and institutional investors operating with Russian assets.

The present study has several limitations that define directions for future research. First, the use of IMOEX as a price-only market proxy introduces a systematic downward bias in the market factor of approximately $0.6\text{--}1.0\%$ per month, as dividend income is excluded from both the index and individual stock returns. While this does not affect the GRS test or cross-sectional pricing results, it precludes direct comparison of the estimated market premium with studies using total return indices; future work should employ dividend-adjusted return series. Second, the sample bias toward liquid, longer-traded securities limits the representativeness of results with respect to small and low-liquidity issuers — a standard constraint in emerging markets research (Foye, 2018; Mosoeu & Kodongo, 2022) that future work may address by extending the observation window as new data become available. Third, the use of size and B/M quintiles as test portfolios partially overlaps with the factor construction sorts; a more stringent robustness check would employ test

portfolios sorted on characteristics orthogonal to factor construction — such as momentum or profitability quintiles — which remains a natural extension of the present work. Fourth, the analysis does not include a momentum factor, which has been shown to carry a premium in a number of emerging markets (Cakici et al., 2013); incorporating momentum into both the factor model and the test portfolio design would allow a more complete assessment of return predictability on the Russian market and constitutes a priority direction for future research.

References

1. Banz, R.W. (1981). The relationship between return and market value of common stocks. *Journal of Financial Economics*, 9(1), 3–18.
2. Cakici, N., Fabozzi, F.J. and Tan, S. (2013). Size, value, and momentum in emerging market stock returns. *Emerging Markets Review*, 16, 46–65.
3. Fama, E.F. and French, K.R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33, 3–56.
4. Fama, E.F. and French, K.R. (1996). Multifactor explanations of asset pricing anomalies. *Journal of Finance*, 51(1), 55–84.
5. Fama, E.F. and French, K.R. (2004). The Capital Asset Pricing Model: Theory and Evidence. *Journal of Economic Perspectives*, 18(3), 25–46.
6. Fama, E.F. and French, K.R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.
7. Fama, E.F. and French, K.R. (2017). International tests of a five-factor asset pricing model. *Journal of Financial Economics*, 123(3), 441–463. doi: 10.1016/j.jfineco.2016.11.004
8. Foye, J. (2018). A comprehensive test of the Fama-French five-factor model in emerging markets. *Emerging Markets Review*, 37, 199–222. doi: 10.1016/j.ememar.2018.09.002
9. Gibbons, M.R., Ross, S.A. and Shanken, J. (1989). A test of the efficiency of a given portfolio. *Econometrica*, 57(5), 1121–1152.
10. Jegadeesh, N., Noh, J., Pukthuanthong, K., Roll, R.W. and Wang, J.L. (2020). Empirical tests of asset pricing models with individual assets. *Journal of Financial Economics*, 138(1), 1–23. doi: 10.1016/j.jfineco.2020.04.008
11. Lintner, J. (1965). The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets. *Review of Economics and Statistics*, 47(1), 13–37.
12. Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77–91. doi: 10.1111/j.1540-6261.1952.tb01525.x
13. Miller, M.H. and Modigliani, F. (1961). Dividend policy, growth, and the valuation of shares. *The Journal of Business*, 34(4), 411–433.
14. Mishra, D.R. and O'Brien, T.J. (2019). Fama-French, CAPM, and implied cost of equity. *Journal of Economics and Business*, 101, 73–85. doi: 10.1016/j.jeconbus.2018.05.002
15. Mosoeu, S. and Kodongo, O. (2022). The Fama-French five-factor model and emerging market equity returns. *The Quarterly Review of Economics and Finance*, 85, 55–76. doi: 10.1016/j.qref.2020.10.023

16. Rosenberg, B., Reid, K. and Lanstein, R. (1985). Persuasive evidence of market inefficiency. *Journal of Portfolio Management*, 11(3), 11–16.
17. Sharpe, W.F. (1964). Capital asset prices: A theory of market equilibrium under conditions of risk. *Journal of Finance*, 19(3), 425–442. doi: 10.1111/j.1540-6261.1964.tb02865.x
18. Zaremba, A. and Czapkiewicz, A. (2017). Digesting anomalies in emerging European markets: A comparison of factor pricing models. *Emerging Markets Review*, 31, 1–15.

Appendix 1

Data collection and preprocessing (Python / Jupyter Notebook)

https://github.com/prokofievtimofei/course-work-EMPIRICAL-TESTING-OF-THE-FAMA-FRENCH-MODEL-USING-RUSSIAN-STOCK-MARKET-DATA/blob/main/data_collection.ipynb

Statistical analysis and factor construction (R)

https://github.com/prokofievtimofei/course-work-EMPIRICAL-TESTING-OF-THE-FAMA-FRENCH-MODEL-USING-RUSSIAN-STOCK-MARKET-DATA/blob/main/final_FF5_code.R